

Fig. 3: Screenshot of sample client application (database)

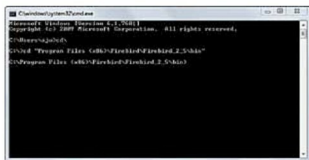


Fig. 4: Firebird path



Fig. 5: Client application testing window

Server.java' to the above registry in the name 'ESoftService.MobileServer' in 'ESoftClient.java' in the client application to point to these services in the corresponding port. Note that in larger projects, we need to write many services for different purposes. So we usually categorise them with a service name. Here we used 'ESoftService.MobileServer' as the service name.

First, we need to initialise the database for 3-tier access using the 'public void setConnection(final ESQlConnection connection, final DB-Connectiondbc)' in the ESoftServer class. There are three ESQlQueries and EDatasetProviders. Each one

of them will be connected to the corresponding database when this method is called. Also, the corresponding provider is added to the 'ESoftServer_Impl' class of the ESoftDBExpress datasnap package, so that these dataset providers can be accessed

at the client side.

Client sample application

A sample client application (ESoftClient.java) is developed and included in the 'com.esoft.Examples' package in this project. The client application is fully designed using Windows Builder plug-in of Eclipse (refer Fig. 2)

The client application is connected to the server with the following code:

```
try {
    final Registry registry = LocateRegistry.
    getRegistry("localhost", 8092);
    System.out.println(registry.list()
    [0]);
    serverIntf = (ESoftServer_Intf)
    registry.lookup("ESoftService.
    MobileServer");
    serverIntf.Authenticate
    ("123654789123654789");
    System.out.println(serverIntf.
    getTestCode());
    query.setCommandText("SELECT
    USERNAME, USERFULLNAME, USERID,
    USERIMAGE FROM USERMASTER");
    } catch (final Exception e) {
```

```
// TODO: handle exception
e.printStackTrace();
}
```

To run both server and client in one system, 'localhost' is used. The port was set to 8092 in the server, so the same port is set in the client side. An authentication method is used in the 'ESoftDBExpressDatasnapESoftServer_Impl' class for security. The security code is set to '123654789123654789'. You can modify the code if you want.

Testing

After installing the required tools as mentioned earlier, run Eclipse.

1. Import the 'Portable Application Server.zip' file included in the EFY DVD into Eclipse. For that go to 'File->Import->General->Existing Project into Workspace' and browse the 'Portable Application Server.zip' file.

You can just put the 'ESoftDBExpress.jar' file anywhere in the project folder, say, 'res' folder. Then refresh the project from Eclipse and add the 'ESoftDBExpress.jar' file to project.

2. For this, from Eclipse, right click on the 'ESoftDBExpress.jar' file and go to 'Build Path' option, select 'Configure Build Path', libraries and click 'Add JAR' file.

3. Firebird 2.5 is used for testing. Firebird is an open source software that can be downloaded from the Internet.

Firebird database 'ESoftECA.rar' file is included in EFY DVD. Install Firebird and restart your system.

4. Now you need to set the password for authentication. In Windows 7, go to 'Desktop->run->cmd->Firebird->bin' directory as shown in screenshot Fig. 4. For Windows XP, go to 'C:\Program Files\Firebird\Firebird_2_5\bin'.

5. From there type 'Gsec -user sysdba -password masterke' and hit 'Enter' key. Then close the window.

6. Now go to project's 'src' folder from Eclipse workspace and open the 'messages.properties' file and set the password as 'masterke' in the code line 'FirebirdPassword=masterke'.